(Please write your Enrollment Number) Enrollment No. 06011720224.

MID-TERM EXAMINATION
(B.Tech CSE-AI) 6th semester
(March, 2024) OFF LINE mode

Subject Code: BAI-304	Subject: Deep Learning-II
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1 (2.5*4)

(a) How does the too high and too low values of learning rate affect the training of the Neural Network?

(b) Why is ReLU the most commonly used Activation Function in DL?

(c) Explain the vanishing gradient problem? What is the formulation of an activation function that predicts the probabilities for n classes (p_1, p_2,..,p_n) such that any predicted probability will never be zero?

(d) Is it beneficial to initialize all the weights with same value of ANN along with dropout layer? Justify

Q2 (Attempt any Two Parts) UNIT-1 (5,5)

(a) Consider a single neuron that takes two inputs (w_1, w_2) which are initialized as (0, 0) with thresholding value as 0.5. What will be the values of (w_1, w_2) after two iterations of perceptron learning algorithm with following data samples:

Sr. No.	X1	X2	Label
1	3	6	N
2	9	4	P
3	2	7	P
4	7	2	N
5	1	7	P
6	8	2	N

(b) What is the effect of low bias and high variance on a deep learning model? Explain the approaches to tackle this scenario.

(c) (i) Suppose you are using early stopping, at which point will the neural network model stop training? Justify.

Sr. No.	Training Loss	Validation Loss
1	3.2	6.9
2	2.9	4.6
3	2.3	2.7
4	1.7	2.5
5	1.2	3.7
6	0.8	4.2

(ii) Explain the different loss functions for regression and classification problems?

Q3 (Attempt any Two Parts) UNIT-2 (5,5)

(a) Explain the Stochastic Gradient Descent, Batch Gradient Descent, and Mini-Batch Gradient Descent methods. What is difference values of epoch and iteration parameters during these gradient descent algorithms?

(b) A four hidden-layer neural network is used for binary classification problem with gradient descent as an optimizer. There are 20 neurons in each hidden layer with input layer of 10 neurons. Explain the effect of using RMSprop and dropout inplace of gradient descent?

(c) Explain the different steps of gradient descent methods along with any two of the following methods:
(i) Adam optimizer,
(ii) Gradient Descent with momentum,
(iii) RMSprop